Chronic Kidney Disease (CKD) Prediction Using Machine Learning

1. Project Overview

The hospital management team requires a predictive model to identify the likelihood of Chronic Kidney Disease (CKD) in patients based on various medical parameters. A dataset containing patient information was provided to develop and evaluate a machine learning model that can classify patients into the appropriate categories.

The primary objective of this project is to develop a reliable classification model that supports the early identification of CKD and assists healthcare professionals in making informed decisions.

2. Machine Learning Approach

This project uses **Supervised Machine Learning**, specifically a **Classification** approach.

Supervised learning is suitable because the dataset contains input features along with known target labels. The model learns patterns from the labeled training data and uses those patterns to predict whether a patient belongs to the CKD or non-CKD class.

3. Dataset Description

The dataset consists of **399 rows and 25 columns**, containing patient records and relevant medical parameters.

The input features are used to train the model, while the target variable represents the patient's CKD classification. The dataset is divided into training and testing sets to evaluate the model's predictive performance on unseen records.

4. Data Preprocessing

One-Hot Encoding is used to convert categorical variables into a numerical format suitable for machine learning algorithms.

This technique represents categorical values as separate binary columns. For example, a categorical variable containing the values *Yes* and *No* can be represented using binary indicators:

Original value	Yes	No
Yes	1	0
No	0	1

This transformation enables the model to process categorical information effectively. Depending on the dataset, additional preprocessing steps, such as handling missing values and preparing numerical features, may also be necessary.

5. Model Selection and Evaluation

Logistic Regression was selected as the best-performing model based on the evaluation results obtained during the model comparison process.

Logistic Regression is a supervised classification algorithm that estimates the probability of an observation belonging to a particular class. In this project, it is used to predict whether a patient is classified as having CKD or not having CKD.

The model's performance is assessed using the following evaluation metrics:

- **Accuracy:** Measures the proportion of correct predictions among all predictions.
- **Precision:** Measures how many predicted positive cases are actually positive.
- **Recall (Sensitivity):** Measures how many actual positive cases are correctly identified.
- **F1-score:** Provides a balance between precision and recall.
- **ROC-AUC score:** Measures the model's ability to distinguish between the two classes across different classification thresholds.

- **Confusion Matrix:** Summarizes correct and incorrect predictions for each class.

For a medical classification task such as CKD prediction, recall for the CKD-positive class is particularly important because false-negative predictions may result in patients with the disease being incorrectly classified as not having CKD.

```
print("The report:\n",clf_report)
```

```
The report:
               precision    recall  f1-score   support

       False       0.98      1.00      0.99        51
        True       1.00      0.99      0.99        82

    accuracy                           0.99       133
   macro avg       0.99      0.99      0.99       133
weighted avg       0.99      0.99      0.99       133
```

```
from sklearn.metrics import roc_auc_score

roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

```
Future_Prediction = grid.predict(X_test[[4]])

print("Future Prediction:", Future_Prediction)
```

```
Future Prediction: [ True]
```

```
Future_Prediction = grid.predict(X_test[[2]])

print("Future Prediction:", Future_Prediction)
```

```
Future Prediction: [False]
```

6. Conclusion

The project demonstrates the application of supervised machine learning classification techniques to Chronic Kidney Disease prediction. The dataset is preprocessed using One-Hot Encoding, and Logistic Regression is selected based on its observed evaluation performance.